

Table 2.2 Spread Cost Analysis

Category	Payment	Study
Order Processing	Order processing fee	Tinic (1972)
	Service provided	Demetz (1968)
Inventory Cost	Risk-reward for holding inventory	Garmen (1976)
		Stoll (1978)
		Ho and Stoll (1981)
		Madhavan and Sofianos (1998)
		Amidhud and Mendelson (1980)
		Treynor (1971)
		Copeland and Galai (1983)
		Gosten and Harris (1988)
		Huang and Stoll (1997)
		Easley and O'Hara (1982, 1987)
		Kyle (1985)
Adverse Selection	Payment for transacting with informed investors	
	Payment for transacting with investor with private information	

Some of the more important research papers for various transaction cost topics and algorithmic trading needs are shown in the following tables. [Table 2.2](#) provides important insight into spread cost components. These papers provide the fundamental needs for order placement and smart order routing logic. [Table 2.3](#) provides important findings pertaining to trade cost measurement and trading cost analysis. These papers provide a framework for measuring, evaluating, and comparing trading costs across brokers, trading, and algorithms. [Table 2.4](#) provides insight into pre-trade cost estimation. These papers have provided the groundwork for trading cost and market impact models across multi-assets as well as providing insight into developing dynamic algorithmic trading rules. Finally, [Table 2.5](#) provides an overview of different trade cost optimization techniques for algorithmic trading and portfolio construction.

THE NEW MARKET STRUCTURE

Over the last few years, there have been some dramatic and significant changes to the structure of the equities markets. Some of the changes were in response to changing regulations and the electronic environment, while some have been due to changing investor preference and investment style. We are certain these will continue to evolve over time.